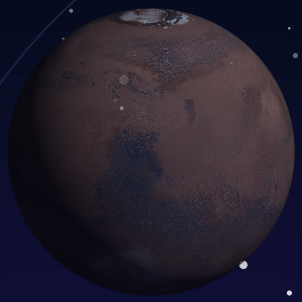
NASA SPACE APPS CHALLENGE 2026 · CAIRO

ELSEWHERE

Exploring other worlds starts with understanding our own.

Project report · Challenge: Identify Earth Locations that Analog the Permanent Moon Base Locations and Mars







TEAM

NOCTRA

- **Alaa Ali**
  Team Leader · Frontend, visualisation & integration
- **Rahma Mohamed**
  Terrain analysis & search engine
- **Randa Ashraf**
  Backend, API & grounded AI explanations

14 November 2026 · https://github.com/Alaa-Alyy/ELSEWHERE

Planet renderings from NASA textures (GIBS Blue Marble; Solar System Treks Viking MDIM 2.1 and LRO WAC) — illustrative, not to scale.

1 At a glance

ELSEWHERE is a screening tool for finding Earth terrain that resembles any place on the Moon or Mars. A user pins a location on a 3D globe. The 20 × 20 km window around it is measured from NASA elevation data and compared with 1,143,707 Earth land windows using one fixed, explainable distance. The closest windows, at least 100 km apart, are returned with the evidence behind each one.

1,143,707	10	0.1–0.4 s	0
Earth land windows searched, 56°S–60°N	numbers in each terrain fingerprint	whole-Earth search per query	trained models: one fixed, declared distance

Key finding: shape and environment are different questions. For Jezero crater (Mars), the closest terrain under the slope-focused profile is in **Montana, USA** (index 96.98), a wet site (636 mm/yr, 4 °C mean). Under the balanced profile, an **arid window in Algeria** (212 mm/yr) ranks first (96.46). ELSEWHERE shows this gap instead of hiding it in one score.

Honesty statement. The terrain similarity index is a heuristic, not a probability or a percentage match. Results are leads to investigate, not validated analog sites, first discoveries or landing-safety assessments. Earth cannot reproduce reduced gravity, vacuum or a thin CO_2 atmosphere, or the radiation environment.

The challenge

Moon and Mars missions test rovers, instruments and crew procedures on Earth first. Classic analog sites (Atacama, Death Valley, Iceland, Hawaii, Antarctica) are well known, but each resembles another world in only some ways. The challenge asks for a *systematic, global* search that looks beyond famous sites, explains what is similar, how similar and what is different, and respects Earth's limits.

Our answer

One pipeline describes every place, on any body, with the same ten-number terrain fingerprint and ranks Earth windows by a declared distance (Figure 1). We keep four kinds of information separate: **measured** terrain, the **computed** index, **context** (climate, planet facts, imagery) and **AI-generated** text.

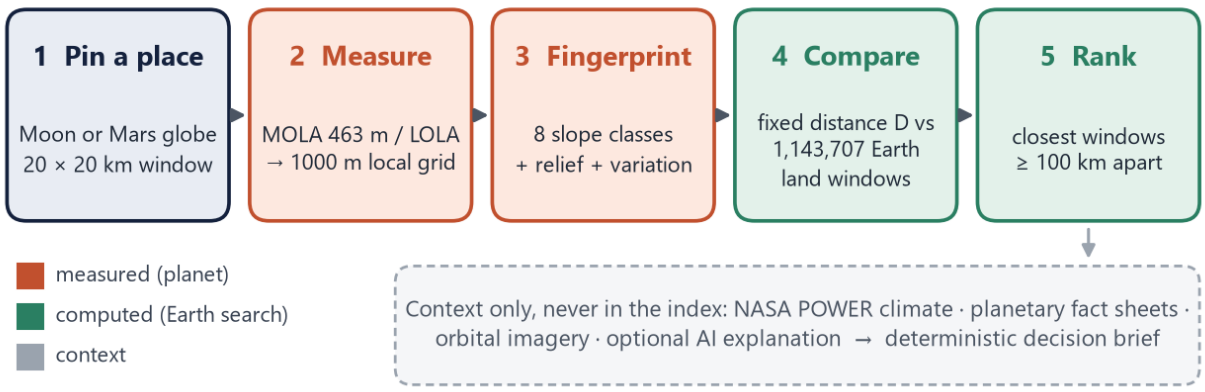


Figure 1. How ELSEWHERE works. Steps 1–5 produce the terrain index; the dashed box lists context that is shown but never scored. (Diagram.)

2 Data and method

Use	NASA dataset	Resolution	Reference
Earth · global search	SRTMGL30 v002 (27 tiles)	~1 km	doi:10.5067/MEaSUREs/SRTM/SRTMGL30.002 · 56°S–60°N
Earth · curated mode	SRTMGL1 v003 (3 regions)	30 m	doi:10.5067/MEaSUREs/SRTM/SRTMGL1.003
Mars	MOLA MGS DEM (NASA/USGS)	463 m	Smith et al. 2001, JGR 106(E10)
Moon	LOLA LDEM_64 · LDEM_80S_80MPP_ADJ (south of 80.5°S)	~474 m · 80 m	Smith et al. 2010; Barker et al. 2023, doi:10.3847/PSJ/acf3e1
Context only	POWER Climatology v2.10.1 · Planetary Fact Sheets · GIBS / Solar System Treks imagery	~50 km · body-wide	never used in the match

Table 1. Open NASA data only. Every source file has a SHA-256 record in `data/sources/`; the global corpus is versioned as `sha256:e483cf61…fe75a`.

Windows on each body's own figure

Each place is a 20 × 20 km window on a 1000 m grid. Elevation is projected to a local azimuthal equidistant grid on the body's own figure (Moon sphere 1737.4 km, Mars sphere 3396.19 km, Earth WGS84) and area-averaged. Earth is tiled into **53,288 parents** of 100 km on a 90 km lattice, with 81 windows per parent at a 10 km stride. A window is kept only if at least 95% of its cells are valid (at least 100 cells); windows with more than 5% cells at exactly 0 m are rejected as likely water.

Fingerprint, distance and index

Part	Definition
Slope distribution (8)	3 × 3 Horn (1981) slope; share of cells in 0–1, 1–3, 3–5, 5–10, 10–20, 20–30, 30–45 and 45–90°
Relief (1)	P95 − P5 of elevation (m)
Local height variation (1)	median |h − mean of 3 × 3 neighbourhood| (m): kilometre-scale, not rock roughness

$$d_{slope} = 0.5 \times \Sigma|p_i - q_i| \quad (0 = \text{identical}, 1 = \text{disjoint})$$
$$d_{relief} = |r_e - r_t| \,/\, \max(|r_e|, |r_t|, 1\ \text{m})$$ $d_{variation}$ likewise
$$D = w \times d_{slope} + (1 - w)/2 \times (d_{relief} + d_{variation})$$
$$\text{index} = 100 \times (1 - D)$$ a heuristic similarity index, not a probability

Profiles: terrain_general w = 0.40 (relief and variation 0.30 each) · slope_focus w = 0.70 (0.15 each) · slider: any w in [0, 1]. Each part lies in [0, 1], so no scaler is needed.

Ranking, and why there is no machine-learning model

scikit-learn `NearestNeighbors` (brute force, callable distance) ranks all windows. Results must be at least 100 km apart (global, top 5) or 20 km apart (curated, top 3). No labelled set of "this Earth place is an analog of that Mars place" exists, so a trained model would only learn our own assumptions. A fixed distance can be explained metric by metric, and users can change the weights and watch the ranking respond.

Context layers (never scored)

Layer	Rule
Climate (NASA POWER, ~50 km)	arid < 250 mm/yr · semi-arid 250–500 · wet ≥ 500 ("likely vegetated"; precipitation is only a proxy)
Mountain caveat	relief > 1000 m and 250–1500 mm/yr → dryness shown as "uncertain"
Scorecard bands	per distance: ≤ 0.05 very similar · ≤ 0.15 similar · ≤ 0.30 partly similar · above: different (a reading convention, not a test)
Planet facts	Mars 0.38 g, ~6.4 mbar CO_2, 24.7 h sol, about −65 °C, 586 W/m² · Moon 0.17 g, near vacuum, 708.7 h solar day, −173 to 127 °C, 1361 W/m² · Earth 1 g, 1014 mbar, 24.0 h

3 Verification and results

All results are real runs on the current corpus; exact rankings depend on the site and profile. With no ground truth for "analog-ness", verification targets consistency, resolution bias and sanity.

Check	Result
Pipeline consistency	A Jezero pin reproduces the stored reference exactly (relief 723.0 m, local variation 4.59 m); a Moon south-pole pin reproduces 2296.1 m / 23.15 m.
Resolution: ~1 km vs 30 m	Same 81 Harrat Khaybar windows, SRTMGL30 vs SRTMGL1: slope distance median 0.012; relief ratio 0.996 (r = 1.0); local variation about 20% lower with ~1 km data (r = 0.99). The UI states this bias.
Positive control (qualitative)	From the lunar south pole (terrain_general), #1 (26.50°S, 66.99°W) falls in our labelled Atacama area (350 km radius; the window is in the Puna highlands) and #4 is in Death Valley. Both are known analog areas, found with no prior knowledge. Encouraging, but not an accuracy measure.

Table 2. Verification checks.

Key finding: terrain versus environment

For Jezero, Montana leads under slope_focus (96.98), but it is wet (636 mm/yr, 4 °C). Algeria is second (95.98) and arid (212 mm/yr), and it leads under terrain_general (96.46). From the lunar south pole, slope_focus returns an Amazon window first (0.75°N, 65.98°W, 2560 mm/yr), a terrain analog only. We also found and handled a data caveat: a Puna window reads 749 mm/yr on the ~50 km POWER grid while the true Atacama core reads 7 mm/yr, which is why the mountain "uncertain" flag exists.

#	slope_focus (w = 0.70)	Index	terrain_general (w = 0.40)	Index
1	**Montana, USA** · 47.96°N, 112.59°W	96.98	**Algeria** · 32.94°N, 0.37°W	96.46
2	**Algeria** · 32.94°N, 0.37°W	95.98	37.26°N, 69.57°E	95.84
3	10.41°S, 27.53°E	95.55	18.14°S, 68.58°W	95.82
4	10.13°S, 32.83°E	95.51	50.96°S, 72.41°W	95.78
5	50.96°S, 72.41°W	95.51	7.22°N, 38.82°E	95.61

Table 3. Top five Earth windows for Jezero under both profiles (≥ 100 km apart), recomputed for this report from the published corpus. Only the two windows discussed are named.

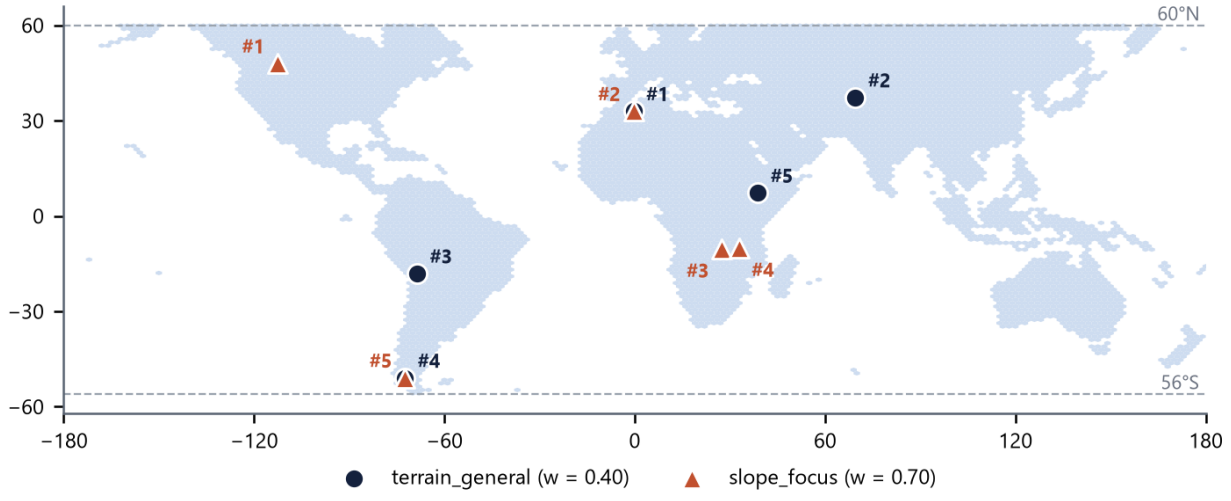


Figure 2. The searched Earth corpus (light blue: cells with valid 20 km windows, 56°S–60°N) and the Jezero results from Table 3.

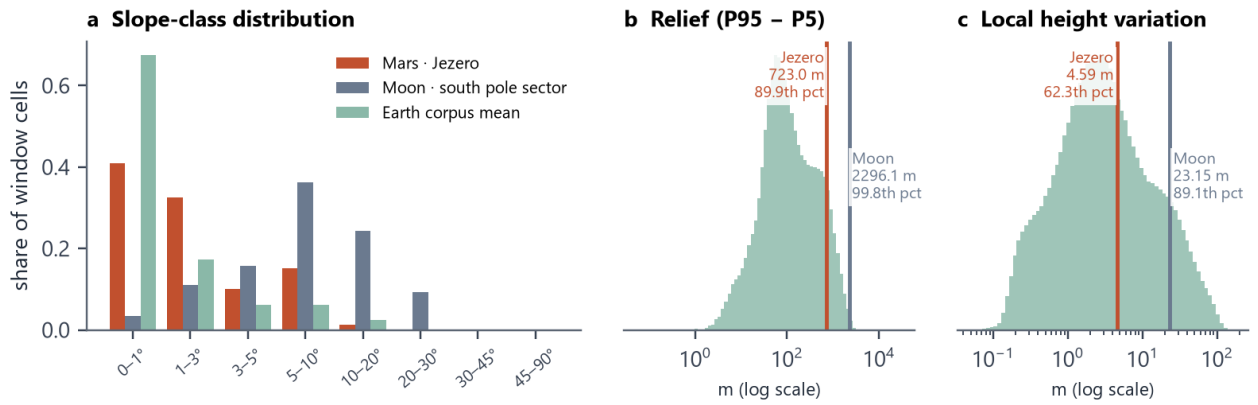


Figure 3. Reference fingerprints. **a** Slope-class shares for Jezero and the lunar south-pole sector, with the Earth corpus mean. **b, c** Relief and local height variation of all Earth windows (log scale), with each reference's percentile, computed for this report from the corpus file.

Why the indices are in the 90s

With 1.14 million candidates and a ten-number fingerprint, close matches almost always exist, so a high index is not evidence of a strong analog. Flat terrain matches most easily: on average 68% of an Earth window's cells lie in the 0–1° class. The lunar south-pole sector, at the 99.8th percentile of Earth relief, has fewer close matches. Nothing is rounded or capped; the UI shows two decimals and explains this in a "Why are scores so high?" note.

4 The ELSEWHERE application

Step	What the user sees
Space hub	A cartoon astronaut; Earth and Mars orbit him, the Moon orbits Earth (NASA textures, not to scale). Click the Moon or Mars and the astronaut jumps onto it.
Experiment	"What experiment will you run?" (rover test drive, landing/moonwalk rehearsal, base-camp survey, field geology). It only selects one of the two fixed profiles.
Pin anywhere	Click a 3D Moon/Mars globe or a named place (Jezero, Gale, Olympus Mons, Hellas, Apollo 11 and 15, Tycho, south pole…). The fingerprint is shown before any search.
Search Earth	A short warp cut-scene while the real search runs (0.1–0.4 s over 1.14 M windows).
Results	The astronaut lands on #1 on a 3D Earth globe and hops between results. Cards show the index (2 decimals), per-metric bars, closest metric and biggest gap, a climate chip and a known-analog label.
Site dossier	Real image pair at the same 60 km scale; 3D blocks from real elevation with equal vertical exaggeration; Similar / Different; research applications with four suitability levels; measurements and sources.
Tradeoffs	Selection vs alternative, metric by metric; a slope slider shows whether the ranking changes.
Decision brief	A deterministic record (ID = SHA-256 of its canonical JSON) with evidence, versions, sources, limitations and a proposed, not executed, test.
Grounded chat	Optional. Google Gemini (gemini-3.1-flash-lite, fallback gemini-3-flash-preview) via the OpenAI-compatible API answers from server evidence only and never computes scores.

Table 4. User journey. Optional CC0 sound (off by default), reduced-motion support, works on phones.

Technology. React 18, Vite 5, CesiumJS, three.js · FastAPI + Pydantic v2 (Python 3.12) · NumPy, SciPy, scikit-learn, pyproj, rasterio (GDAL). **Tests:** 94 Python and 37 frontend tests, all passing.

5 Limitations, impact and next steps

Limitation	What it means
Terrain shape only, 1 km scale	Rocks, regolith, mineralogy, ice/permafrost, dust, biology, caves/lava tubes, polar illumination and land access are not assessed.
Coverage and resolution	Earth search covers 56°S–60°N only (no polar regions); local variation is biased low by about 20%.
Context data	Climate is a ~50 km model (mountain rainfall can be wrong); Mars/Moon rows are planet-wide values, not site measurements.
No ground truth	The index is a heuristic; overlapping windows are correlated; field checks (site visit, drone survey) are needed before any test.
Earth's limits	Reduced gravity, vacuum or thin CO_2 atmosphere and radiation cannot be reproduced.
No discovery claims	"Not in our list of well-known analog sites" is never a first-discovery claim.

Table 5. Limitations stated in the report and in the application.

Impact

A fast, transparent first screening of Earth test sites for rover mobility, autonomous navigation, EVA rehearsal, landing-site training and remote-sensing validation. It is open, reproducible and built on NASA data only, so a mission team can see exactly why a site was proposed and what the comparison leaves out.

Next steps

Geology and mineralogy layers (geologic maps, ASTER) · MODIS NDVI as a vegetation filter · polar Earth coverage (Copernicus DEM, ArcticDEM) · finer DEMs for shortlisted sites · Mars/Moon thermal and illumination data · field validation with analog-site experts.

AI disclosure

In the product: one LLM (Google Gemini via the OpenAI-compatible API) writes optional explanations from server-provided evidence only; it does not compute features, scores or rankings, and the decision brief works without it. **In development:** the team used AI coding assistants (including Anthropic Claude) to draft code, tests and documentation; all results come from the code and data in the repository.

References and data citations

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[11] ELSEWHERE repository (code, SHA-256 data provenance, manifests, tests): https://github.com/Alaa-Alyy/ELSEWHERE